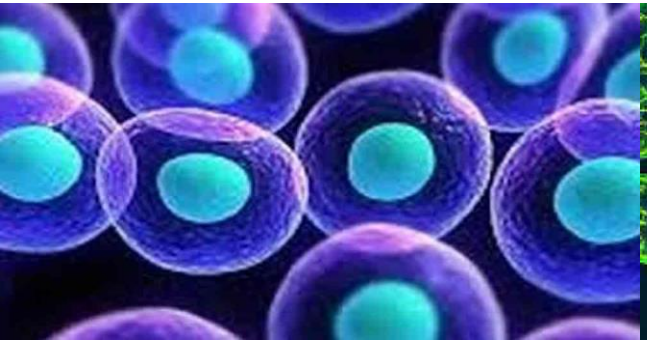
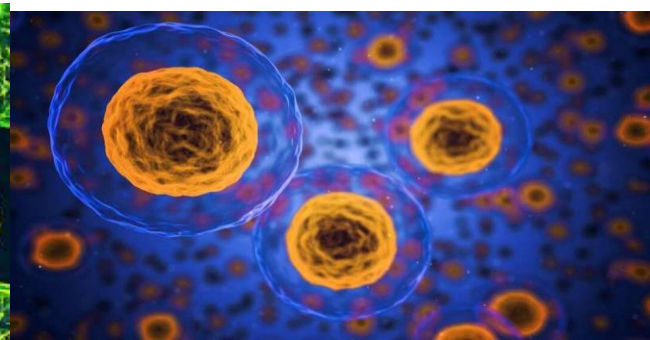
Part 6

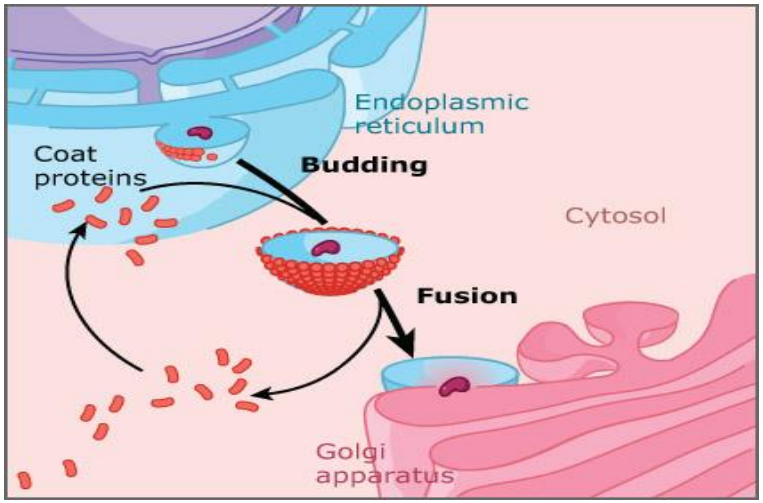
The cytoskeleton



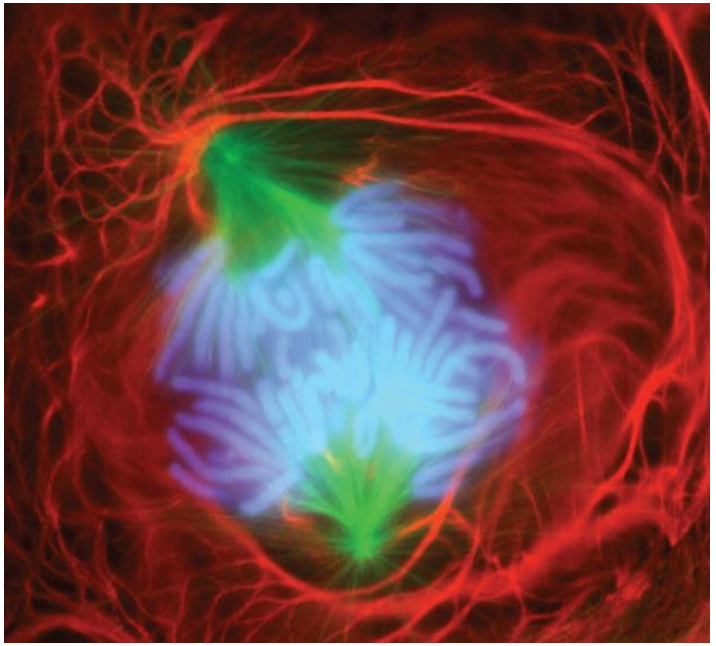


The cytoplasm of eukaryotic cells is in constant motion

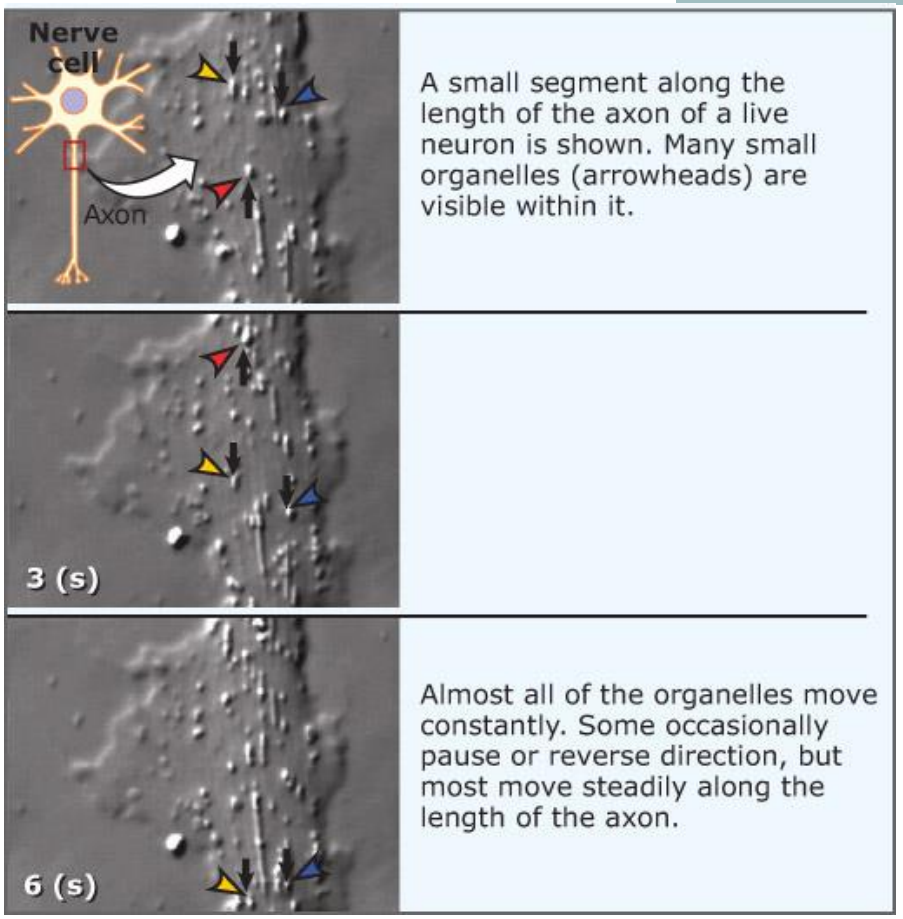
Movement of organelles



Secretory vesicles leave the ER, fuse with Golgi near the centre of the cell, and are transported to the plasma membrane



In mitotic cells, chromosomes first align and then move to opposite sides of the cell

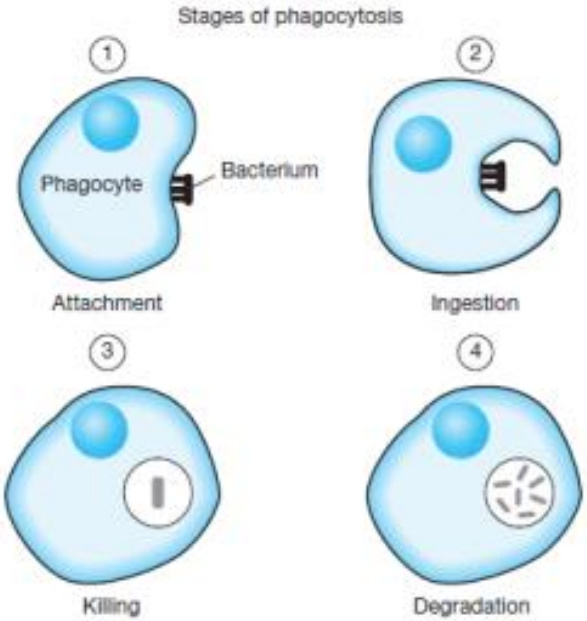


Three images, from a video, show an axon from a living neuron. Three vesicles, marked by red, yellow, and blue arrowheads, are followed over a 6-second interval. Two vesicles move toward the tip of the axon and one moves toward the cell body. (Photos courtesy of Paul Forscher, Yale University.)

Movement of organelles and chromosomes to the right place, at the right time, is accomplished by the **cytoskeleton**

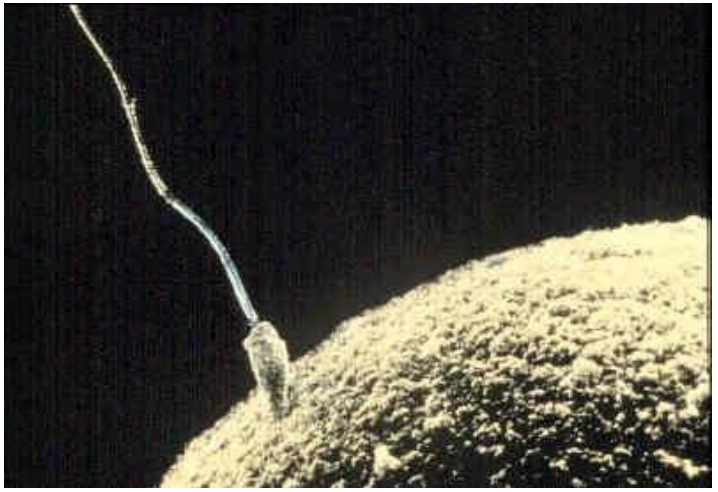
The movements of motile cells

Within body (animal cells)



The white blood cells that track and destroy bacterial invaders crawl across a surface

Through the environment (single celled organism or gametes)



Sperm cells swim through fluid to reach their destination

Cytoskeleton

❖ Function:

- Movement

- Transport

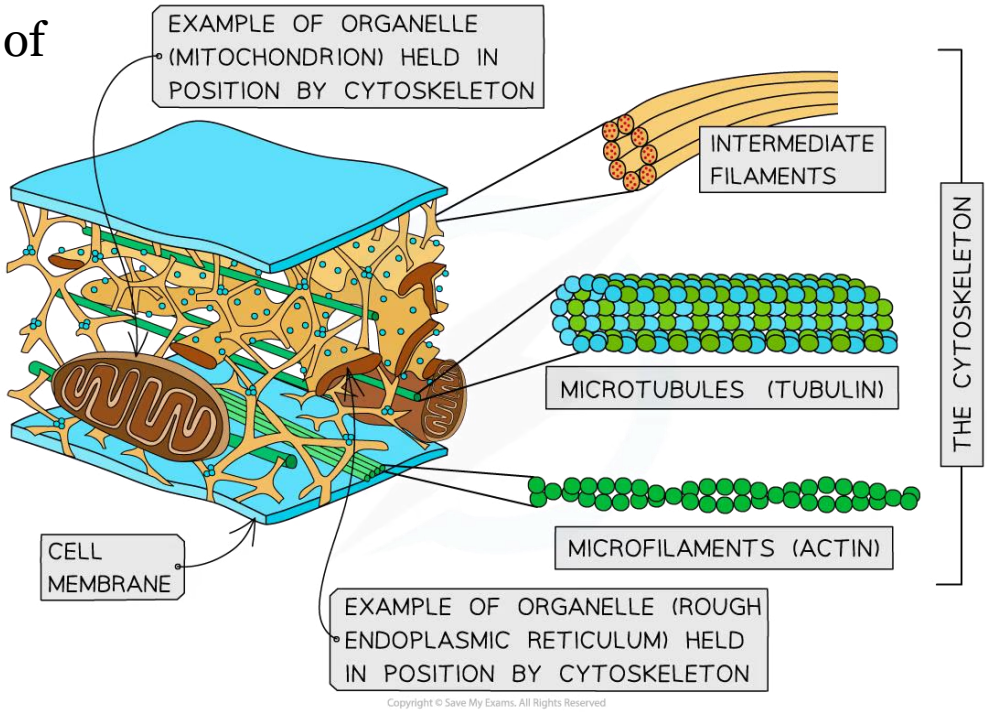
- Organization and structural support that define the shapes of all cells

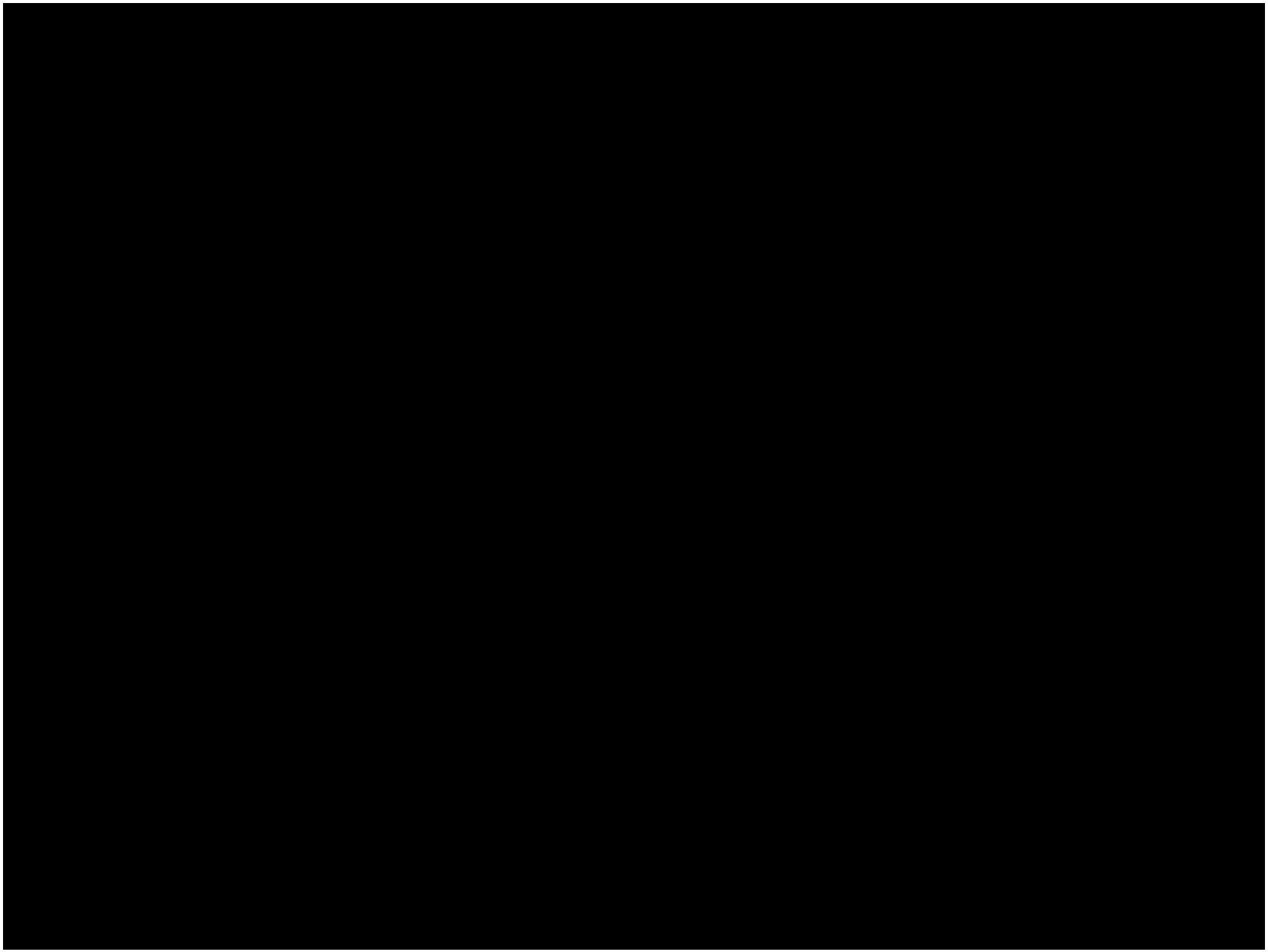
❖ Composed of three major types of structural proteins:

-Microtubules

-Intermediate filaments

-Actin



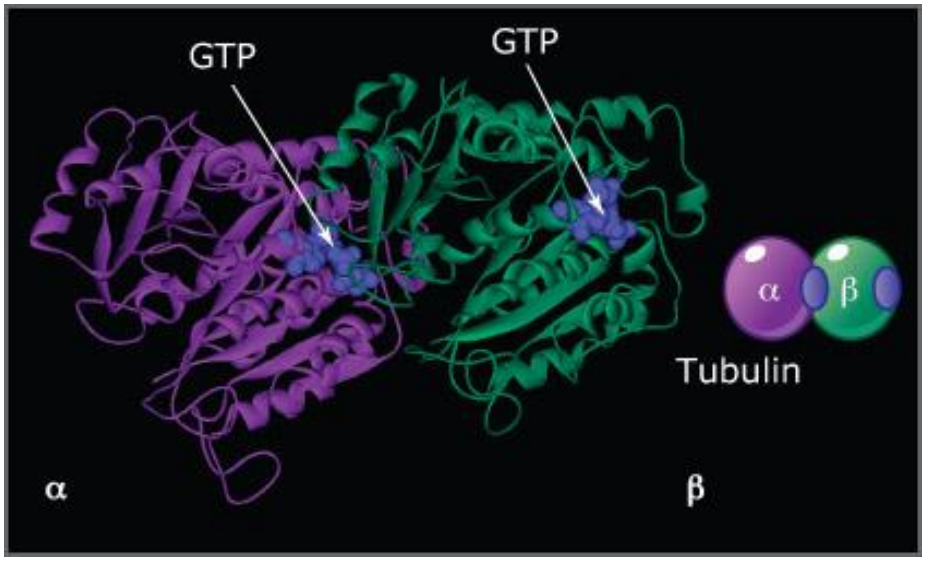


1. Microtubules

❖ The basic subunit that forms a microtubule is the protein **tubulin (**hollow polymers of tubulin heterodimers)

❖ Each molecule of α- and β-tubulin binds a molecule of guanosine triphosphate (GTP), near the front of the tubulin

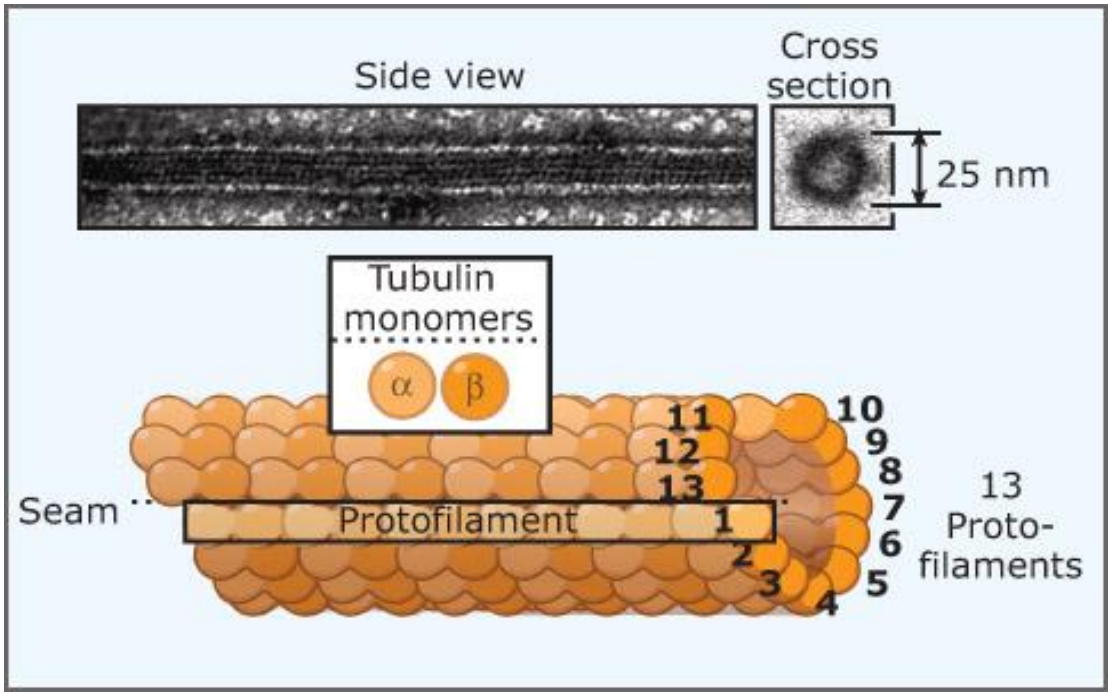
❖Microtubules are cut by **katanin**, a protein named for the Japanese word for sword



The three-dimensional structure of the tubulin heterodimer, the basic building block of a microtubule. In purple and green colors are the polypeptide backbones of the two protein subunits, and in blue are the two molecules of GTP that are bound to each heterodimer. Note the similarity in the structures of the two subunits and that they are arranged head to tail in the complex. At the right is a schematic drawing that shows how the dimer is represented in the figures in the chapter

Structure from Protein Data Bank 1TUB. E. Nogales, S. G. Wolf, and K. H. Downing, Nature 391 (1998): 199-203.

1. Microtubules



The structure of a small segment of a microtubule.
The individual tubulin **heterodimers** are aligned end to end in straight **protofilaments**, and the protofilaments are arranged side by side to form a hollow tube. All heterodimers have the same orientation, with the **β-subunit toward one end** of the microtubule and the **α-subunit toward the other**. At the top are electron micrographs of microtubules assembled from pure tubulin

1. Microtubules

- Mitotic spindle
- Form the core of motile structures: **cilia and flagella**

1. Microtubules

Drugs that make microtubules more or less stable block mitosis and are used to treat cancers

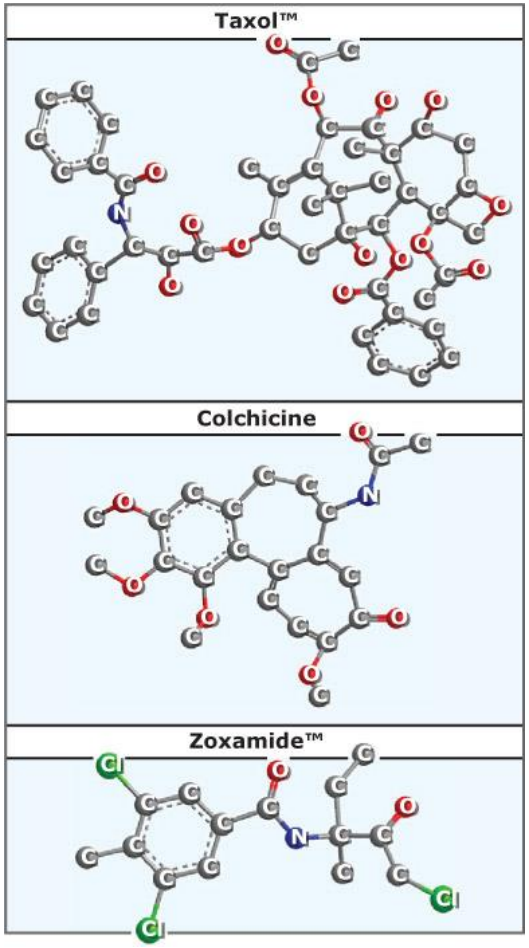
Taxol (paclitaxel) binds to microtubules and makes them more stable by preventing tubulin subunits from dissociating

→ Used to treat ovarian and breast cancers

Colchicine: causing a cell's microtubules to disappear

→ Used to treat gout

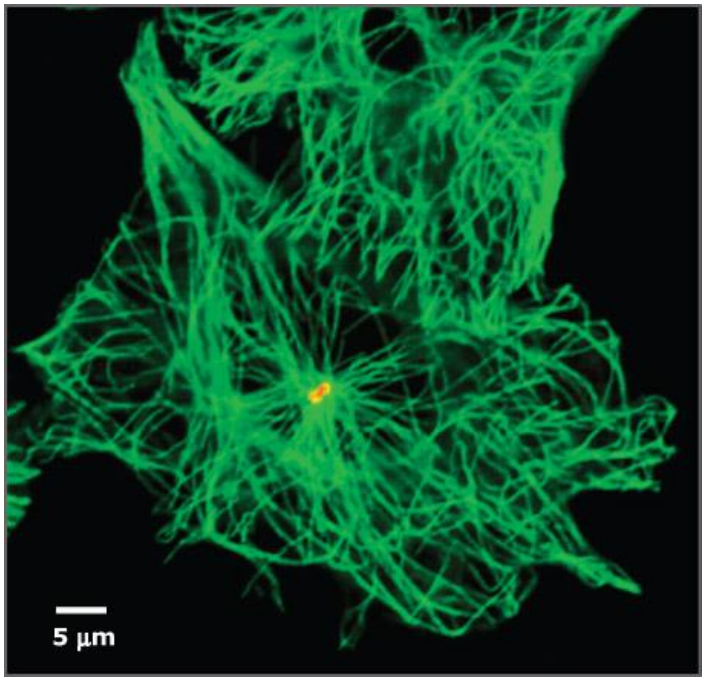
Zoxamide, a fungicide, binds specifically to fungal tubulins and prevents fungal growth



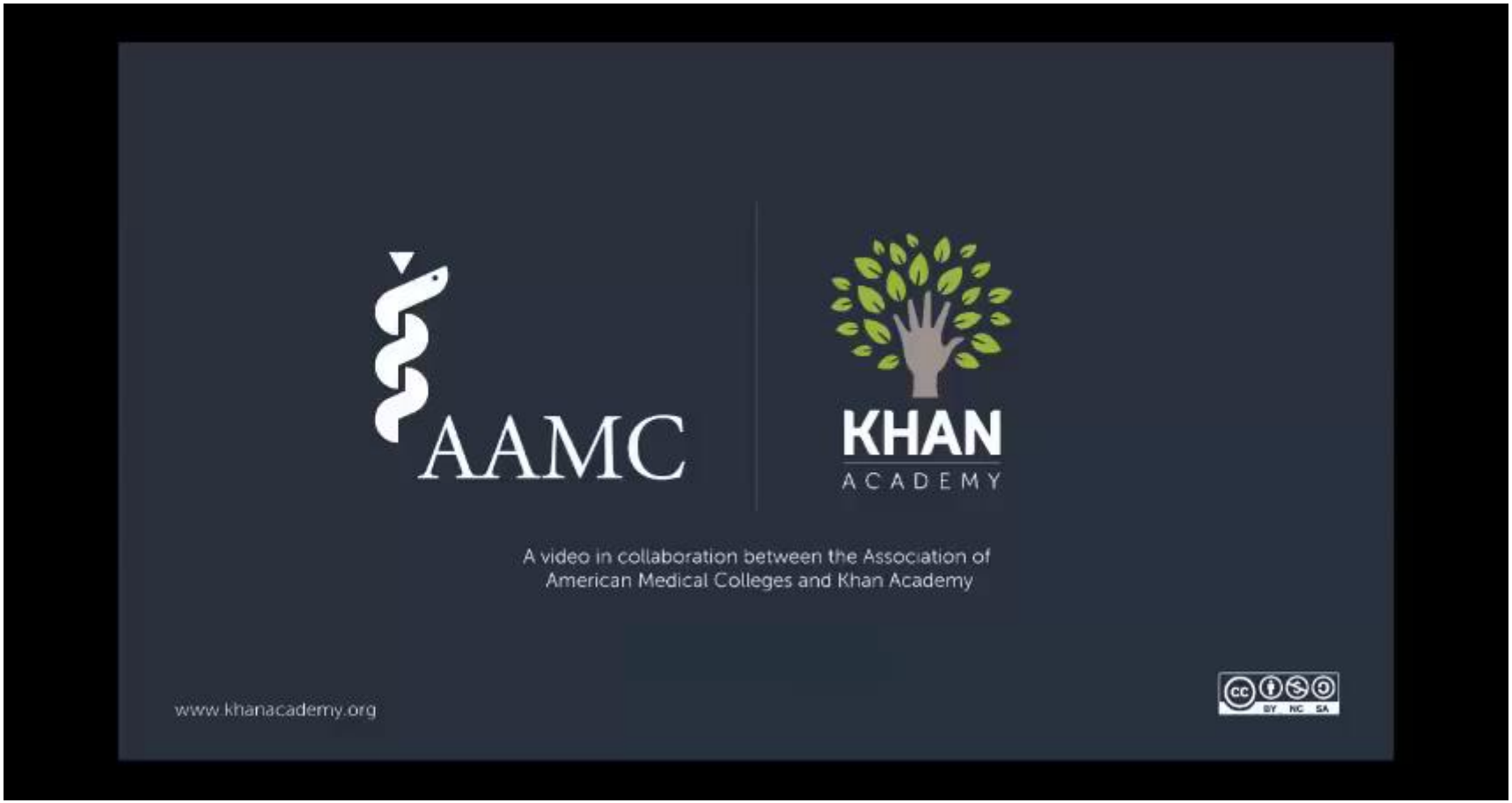
The structures of three small organic molecules that disrupt microtubule assembly or disassembly. Paclitaxel (Taxol™) and colchicine are natural products produced by specific plants (the Pacific yew tree and the meadow saffron, respectively). Zoxamide™ is a manufactured molecule discovered by screening a large collection of small molecules for those with the ability to interfere with microtubules

1. Microtubules

Microtubule organizing center (MTOC):
+Centrosome
+Basal body



The microtubules in a fibroblast cell, viewed after labeling them with a fluorescent dye so that they appear green. The microtubules are organized around a point (red) near the center of the cell and extend throughout the cytoplasm. Most microtubules are long enough to run from one part of the cell to another. (Photo courtesy of Lynne Cassimeris, Lehigh University.)



https://www.khanacademy.org/test-prep/mcat/cells/cytoskeleton/v/microtubules-2

2. Actin

Actin is a ubiquitous and essential protein found in all eukaryotic cells.

Highly conserved; 90% identical between species

Actins filaments make up **microfilament**

Microfilaments function

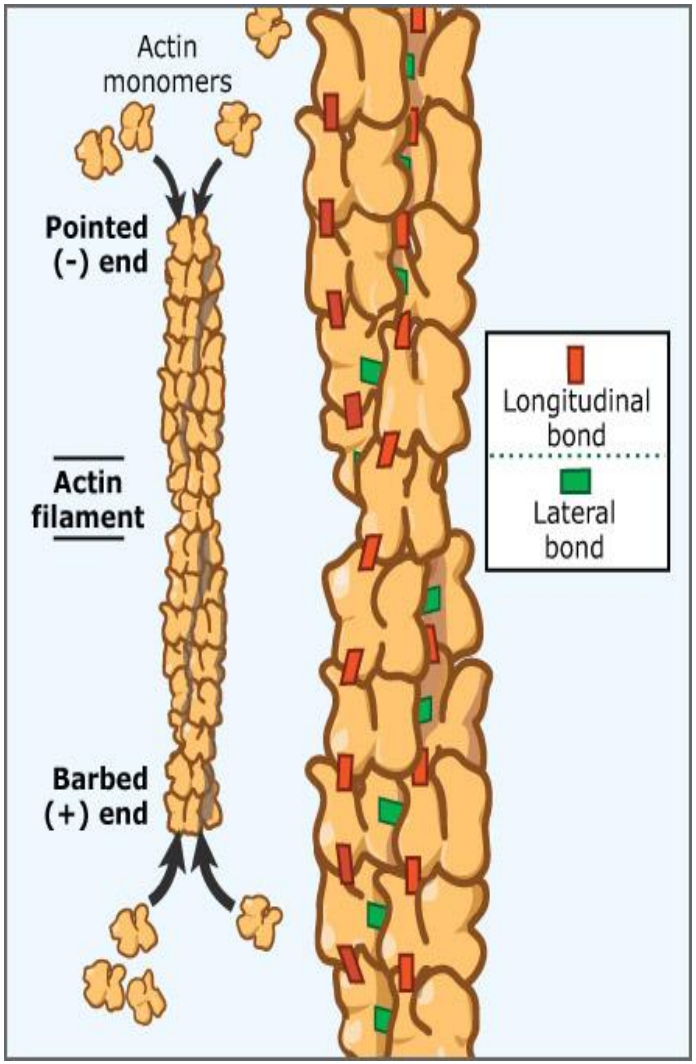
-maintain cell shape
-muscle contraction
-cytoplasmic streaming
-cell motility
-cell division (cytokinesis)

2. Actin

Actin monomers self-associate and polymerize into filaments **~8 nm** in diameter

An actin filament resembles a double-stranded string of beads that has a right-handed helical twist

Dynamic: longer by actin polymerization, shorter by actin depolymerization



Actin monomers assemble head-to-tail into double-stranded, polar filaments.

3. Intermediate filament

Intermediate filaments, or nanofilaments, are major components of the cytoplasmic and nuclear cytoskeletons, made of various type of protein

8–12 nm thick and form resilient networks in the cytoplasm and nucleus.

Essential to maintain correct tissue structure and function.

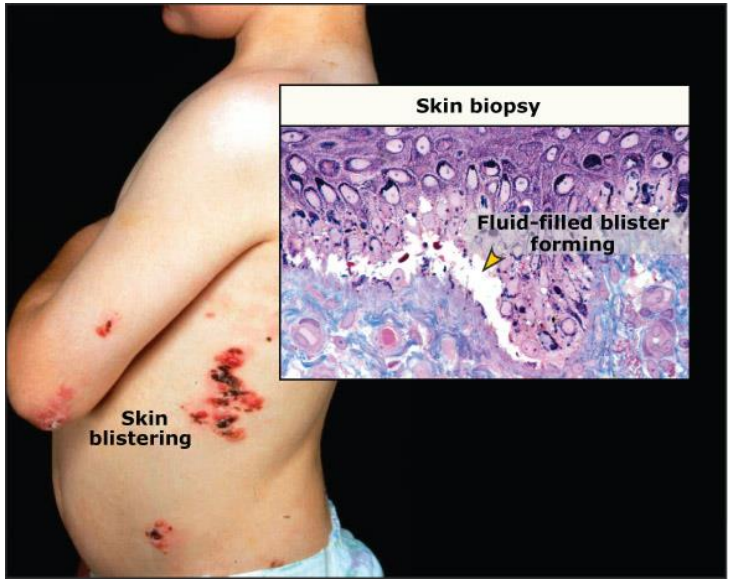
Structural support, resist mechanical stress

Defective intermediate filaments are the cause of many human diseases

3. Intermediate filament

Most intermediate filament genes in humans encode keratins, expressed in all different kinds of epithelial tissues of the body

Mutations in keratins cause epithelial cell fragility, an inherited skin-blistering disorder, namely EBS



The condition illustrated here is EBS, the first disorder linked to intermediate filament gene mutations, caused by mutations in keratins K5 or K14. The main photo shows the characteristic skin blistering caused by scratching, rubbing, or tight clothing. The inset shows a section through a diagnostic skin biopsy taken after rubbing the skin surface with a pencil eraser: The basal layer of epidermal keratinocytes has a fluid-filled blister due to rupturing of the cells

Motor Proteins

Kinesin
These proteins move along the microtubules carrying the cellular components. They pull the organelles along the cell membrane.

Dyneins
These pull the cell organelles towards the nucleus.

Myosin
These interact with actin protein and are responsible for muscle contractions. They also perform cytokinesis, exocytosis, and endocytosis.

